

RESOLUTION NO. _____

ENTITLED: "A RESOLUTION AUTHORIZING SUBMISSION OF AN APPLICATION TO THE WYOMING WATER DEVELOPMENT COMMISSION GROUND WATER EXPLORATORY GRANTS PROGRAM FOR THE BORIE WELLFIELD EXPANSION PROJECT IN THE AMOUNT OF \$400,000.00 AND AUTHORIZING THE DIRECTOR OF THE BOARD OF PUBLIC UTILITIES TO EXECUTE THE APPLICATION AND ALL RELATED DOCUMENTS."

WHEREAS, Wyoming Statute § 15-7-401, provides for a Board of Public Utilities (BOPU) to manage, operate, maintain, and control municipal waterworks, sanitary sewer system and sewage disposal plants;

WHEREAS, Cheyenne City Code, Section 2.68.010, establishes that the BOPU has the exclusive authority to manage, operate, maintain and control the Cheyenne Municipal Waterworks, Sanitary Sewage System and Sewage Disposal Plants of the City and to make all rules and regulations necessary for the safe, economical and efficient operation and maintenance of the above facilities;

WHEREAS, the 2013 Water and Wastewater Master Plan and 2005 Cheyenne Belvoir Ranch Groundwater, Level II, Study recommend the development of additional groundwater resources;

WHEREAS, the City of Cheyenne and BOPU own groundwater resources in the High Plains Aquifer under the eastern portion of Belvoir Ranch (Borie Wellfield);

WHEREAS, the BOPU desires to pursue the Borie Wellfield Expansion Project (PROJECT) to locate, drill, complete, and test two new wells;

WHEREAS, the PROJECT cost estimate for engineering, construction, and other administrative costs is eight hundred thousand dollars (\$800,000), which includes drilling services, construction and miscellaneous costs;

WHEREAS, the PROJECT meets the criteria and is eligible for consideration of grant funding from the Wyoming Water Development Commission (WWDC) Ground Water Exploratory Grants program in the amount of four hundred thousand dollars (\$400,000), fifty percent (50%);

WHEREAS, the BOPU will match funds in the amount of four hundred thousand dollars (\$400,000), fifty percent (50%), from System Development Fees as part of Fiscal Year 2024 Budget; and

WHEREAS, the BOPU recognizes that the PROJECT is necessary and that it is essential to provide additional water supply for the City of Cheyenne, especially during potential Colorado River Curtailment.

NOW, THEREFORE, BE IT RESOLVED BY THE GOVERNING BODY OF THE CITY OF CHEYENNE, that an application be submitted to the Wyoming Water Development Commission for a Ground Water Exploratory Grant in the amount of four hundred thousand dollars (\$400,000.00) for the Borie Wellfield Expansion Project; and

BE IT FURTHER RESOLVED, that the Mayor, the City Clerk, and the Director of the Board of Public Utilities are hereby designated as the authorized representatives of the City of Cheyenne to act on behalf of the City of Cheyenne in all matters related to this application and related documents, including any amendments to the Project Agreement.

PRESENTED, READ AND ADOPTED this _____ day of _____, 2023.

Patrick Collins, Mayor

(SEAL)

ATTEST:

Kristina F. Jones, City Clerk



THE STATE OF WYOMING

Water Development Commission

6920 YELLOWTAIL ROAD TELEPHONE: (307) 777-7626 CHEYENNE, WY 82002



WATER DEVELOPMENT PROGRAM APPLICATION

GROUND WATER EXPLORATORY GRANTS

Grant funding for ground water studies is based on WWDC recommendations and is sourced from block appropriations by the legislature from the Water Development Accounts. The WWDC authorizes ground water grant expenditures as a maximum 75% project cost share to eligible entities before project work can begin. Applications to fund new or continuing groundwater studies **must** be submitted at least **45 days prior** to scheduled WWDC meetings for consideration. Ground water exploration grants are promulgated by statute (W.S. 41-2-119. Groundwater studies), governed by WWDC Rules and Regulations, and managed according to program "Guidelines:" http://wwdc.state.wy.us/groundwater_grant/groundwater_grant_projects.html

Applicants seeking ground water exploration grant funding must be an incorporated municipality, water district, water & sewer district, or service & improvement district with taxing and/or assessment authority. Private corporations, individuals, and other special districts are not eligible for assistance. Use of grant funds is limited to feasibility study of and/or exploration for the use of ground water for municipal and rural domestic purposes. The WWDC grants for groundwater studies are not to exceed \$400,000 for any one study and/or exploration program. The sponsor receiving the grant shall provide at least 25% of the cost of the feasibility study and/or exploration program from its own funds. **The sponsor is solely responsible for acquiring access, easements, or rights-of-way necessary for exploration.** All grants terminate 24 months from the date of the award.

Note: If you are seeking Level II or Level III funding for a ground water source supply project, this is the wrong application. You must complete a Planning or Construction program application for Municipal or Rural Domestic System projects.

APPLICATION REQUIREMENTS

- The person signing the application must have authority to commit the entity to a binding contract.
- A notarized copy of a resolution supporting this application passed by the board or other governing body of the entity must be provided.
- A project area map (8.5" x 11" preferred) showing corporate boundaries, project location and features should be provided. Include any reports or other supporting information available.
- The project sponsor must govern a system that includes a minimum of 15 taps with meters on each tap.

I. ENTITY INFORMATION

City of Cheyenne Board of Public Utilities

1. Applicant – Name of Entity				
Municipality		1943		
2. Type of Entity (Municipality, Joint Powers Board, etc.)				
2416 Snyder Avenue				
3. Year of Formation				
4. Physical Address (Street Address)				
Cheyenne	Laramie	WY	82001	307-637-6460
5. City	6. County	7. State	8. Zip Code	9. Phone

10. Mailing Address (If Different From Above)		
Bryce Dorr	307-432-2618	bdorr@cheyennebopu.org
11. Primary Contact Person (Type or Print Name)	12. Phone	13. Email
Monday - Friday 8:30am - 4:30pm		
14. Best Time(s) to Reach Contact Person		
Frank Strong, Engineering Manager		
15. Authorized Official (Type or Print Name)	16. Signature of Authorized Official	17. Date

Before applying for project funding, the Wyoming Water Development Commission strongly recommends completion of a Public Water System Survey or Irrigation System Survey available through the link on the website located here: <http://wwdc.state.wy.us/surveys/surveys.html>

18. Has the application entity completed this survey? ☒ Yes ☐ No (last completed in 2018)

If the application was prepared by someone other than the contact person, please provide the following:

19. Name	20. Affiliation	21. Phone	22. Email
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II. PURPOSE AND NEED

1. Provide a brief statement describing the project for which funding is sought, and include the reasons the project is needed. Describe the current situation with the water supply that will be improved by the project. What is the factor that is presently limiting the system supply capacity (e.g., treatment, storage, transmission, etc.)? Attach additional information if needed:

The project includes locating, drilling, completing, and testing two well sites. After rating the new wells, they (and two other existing wells) will be connected to the BOPU Borie Wellfield and ultimately to King II storage tank and the Cheyenne distribution system. The additional groundwater supply is needed to supplement BOPU water portfolio in the event of Colorado River Curtailment and a reduction up to 15,000 acre-feet per year of Cheyenne's firm yield supply (68% of total).

2. Is the applicant entity under any federal (EPA) mandates to improve the water system (e.g., administrative orders, violations, actions taken, etc.)?

No

3. Operating Criteria of the Wyoming Water Development Program advocates prioritization of projects that may serve more than one entity or purpose and those that provide water to a larger, more regional area. Is the applicant entity currently served by a regionalized water supply system? If so please specify. Or will the applicant entity consider regional solutions to the purpose and needs of its water supply system?

BOPU does not operate a regionalized system, but does serve consecutive systems (i.e., South Cheyenne Water and Sewer District and Warren Air Force Base). BOPU has planned for potential rural customers to connect to its system in case of failing wells or declining water quality.

4. List any previous work (studies) completed by or for the entity:

Cheyenne Belvoir Ranch Level II Study, December 2005

Belvoir Wells No. 5&6 Pumping Test Report Aquifer Impact Report, October 2007

Belvoir Ranch High Plains Aquifer - White River Study, January 2009

5. Provide a brief description of the water system operations. Identify the public works director. Identify any system automation within the operations (e.g., SCADA) and existence of any hydraulic models of the system. Describe existing mapping (e.g., paper, map books, GIS). Attach additional information if needed:

Trans-basin surface, local surface, and groundwater sources are treated and blended prior to entering the distribution system in Cheyenne. The source of supply, distribution, collection, and treatment systems are all monitored and controlled via SCADA system. Hydrologic and hydraulic models are used to evaluate system performance and plan for improvements/expansion. Source of supply, distribution, and collection infrastructure are inventoried and mapped using GIS. Water treatment and reclamation facilities are currently being integrated into the BOPU GIS.

III. PERTINENT INFORMATION

The intention of this section is to gather information on your existing water supply facilities. Answer all questions as completely and accurately as possible. Do not leave any questions blank. If questions are not applicable to your system, respond, "N/A." If an answer to a question is unknown, respond, "Unknown." If you need help, please call the Water Development Office at 307-777-7626.

1. Existing Water Supply System

A. EPA Public Water System (PWS) Identification Number: 5600011C

B. Groundwater

(1) Number of Wells: 39

(2) Primary Supply Aquifer(s) or Formation(s): Ogallala and White River

(3) Total Average Production Yield of All Wells (GPM): 1,984 gpm

C. Surface Water

(1) Source Name(s): Crow Creek, Douglas Creek, and North Fork Little Snake River

(2) Type of Diversion(s) (Headgate, Infiltration Gallery, Pumps, Etc.):
Headgates, dams, and reservoirs

(3) Total Average Diversion Yield (CFS or GPM): 33.1 cfs

D. Springs

(1) Name of Spring(s): NA

(2) Total Average Production Yield of All Springs (GPM): NA

E. Water Rights

(1) For the water source supply (or supplies) described above, does the applicant entity possess valid and/or adjudicated water rights?

Yes

F. Transmission Pipeline

(1) Maximum Capacity of the Transmission Pipeline(s) (Gallons per Day): 56,000,000 gpd

(2) Increased Capacity Needed (If Known) (Gallons per Day): 2,300,000 gpd (at new wells)

(3) Approximate Distance from Source(s) to Distribution System: 75 miles

(4) Transmission Pipe Diameter(s): 20-inch to 54-inch

(5) Type of Transmission Pipe Material(s): steel, ductile iron, cast iron, concrete

(6) Age of Transmission Pipeline(s): 14-106 years

(7) Condition of Transmission Pipeline(s): Good to Excellent

G. Water Storage

Hog Park 22,650 AF, Rob Roy 35,640 AF, Lake Owen 684 AF, Granite 5,320 AF, Crystal 3,400 AF, New North Crow 98 AF, Old North Crow 98 AF, South Crow 12 AF

(1) Raw (Volume and Tank Description):

(2) Treated (Volume and Tank Description):

King I 5 MG, King II 15 MG, Round Top 11 MG (offline in 2024), Buffalo Ridge 5 MG, Swan Ranch 0.5 MG, North City 6 MG (future)

H. Treatment

(1) Specify Water Treatment (None, Chlorination, Filtration, Etc.): Flocculation, Sedimentation, Filtration, Chlorination

2. Existing Water Distribution System

A. Is the water use metered? Yes B. Are the billings based on meter readings? Yes

C. Identify unmetered usage (e.g., irrigation of parks, cemeteries, fire protection, etc.):

fire hydrant flushing, firefighting, street sweeping, and leaks at approximately 5%

D. Average Day Demand Water Usage (Gallons per Capita per Day): 95 gpcd

E. Maximum Day Demand Water Usage (Gallons per Capita per Day): 247 gpcd

F. Peak Hourly Demand Water Usage (Gallons per Capita per Day): 504 gpcd

G. Distribution Pipe Diameter(s): 4-inch to 42-inch

H. Type of Distribution Pipe Material(s): steel, ductile iron, cast iron, concrete, PVC, HDPE

I. Age of Distribution Pipeline(s): 0-115 years

J. Condition of Distribution Pipeline(s): poor to excellent

K. Estimated System Water Losses (Percentage): 5%

L. Describe any fire flow protection that the system provides:

BOPU system is designed to provide up to 3,500 gpm for 3 hours for fire fighting.

Hydrants and fire lines with fire suppression systems are used.

M. What water conservation measures are employed?

Previously implemented Plan for Wise Water Use including tiered water rates and outdoor watering schedules and best practices. Currently working to implement drought tolerant landscaping for development projects and advanced metering infrastructure to address leaks.

N. Is there an independent raw water irrigation system? Yes

(1) Source: surface water and recycle water

(2) Raw Water System Capacity (Gallons per Day): 1,500,000 gallons surface; 3,000,000 gallons recycle

(3) Average Annual Raw Water Usage (Gallons per Year): 141,000,000 gallons surface; 98,000,000 gallons recycle

3. Demographic Information and Existing Water Service Area

A. Population (2010 Census): 59,466 B. Current Population Estimate: 65,501 (service area 75,000)

C. Does the applicant have a comprehensive planning boundary? Yes

(1) If so, what is the estimated additional population that may be served in the future? 95,000

D. How many taps are served within the service area? 24,787

E. How many taps are served outside of the service area? 16 (Polo Ranch irrigation and stock tanks)

F. Identify names of other water systems served:

Warren Air Force Base, South Cheyenne Water and Sewer District

G. Identify any existing planning reports (municipal or county) that address growth management in the project area. Provide titles and how copies of the reports could be obtained:

Cheyenne Municipal Storage Level II Study (on file WWDO)

4. Financial Information

A. Rates

(1) Tap Fee(s) – Residential: for 3/4", \$9,818.70 (includes tap and system development fees)

(2) Tap Fee(s) – Commercial: for 3/4", \$9,818.70 (includes tap and system development fees)

(3) Average Residential Monthly Water Bill and Corresponding Gallons Used:

\$38 per month at 6,400 gallons usage

(4) Water Rates (Provide rates for all tiers and categories of use. Attach additional pages if needed.):

Attached

(5) Identify any local conditions that affect the water rates (e.g., flow-through for frost prevention, etc.):

None

B. Financial Statement (of Water Utility)

(1) Revenues

a. Annual Revenues Generated from Water Sales:	<u>\$19,572,000</u>
b. Annual Revenues from Tap Fees:	<u>\$ 2,335,000</u>
c. Annual Revenues from Other Sources:	<u>\$14,396,073</u>
d. Total Annual Revenues:	<u>\$36,303,073</u>

(2) Expenditures

a. Annual Budget for Operation and Maintenance Expenses:	<u>\$15,714,023</u>
b. Annual Payments for Debt Retirement:	<u>\$ 2,801,400</u>
c. Annual Payments to a Repair and Replacement Fund:	<u>\$ 0</u>
d. Annual Payments to an Emergency Fund:	<u>\$ 0</u>
e. Annual Payments for Other Purposes:	<u>\$17,787,650</u>
f. Total Annual Payments:	<u>\$36,303,073</u>

(3) Other

a. Balance in Repair and Replacement Fund:	<u>\$ 4,709,388</u>
b. Balance in Emergency Fund:	<u>\$ 5,310,720</u>
c. Annual Cost of Water Quality Testing:	<u>\$ 324,070</u>

(4) Is the operation of the water system self-supporting in terms of revenues offsetting costs for operation, maintenance, debt retirement, replacement funds, emergency funds, etc.? Yes

a. If not, how is the difference subsidized?

NA

C. How do you intend to fund the sponsor cost share to match WWDC grant funds for this project (e.g., special/general purpose tax, other grant or loan funds, fiscal budget expenditure, etc.)? If possible, attach latest fiscal year financial statement that would demonstrate available funds. If a suitable source of supply is found, also inform the WWDC in this synopsis how you would intend to finance subsequent production and transmission systems (e.g., WWDC Level III funding, State Drinking Water Revolving Fund loan, State Land & Investment royalty grant/loan, special/general purpose tax, Federal program grant/loan, local sources, etc.)

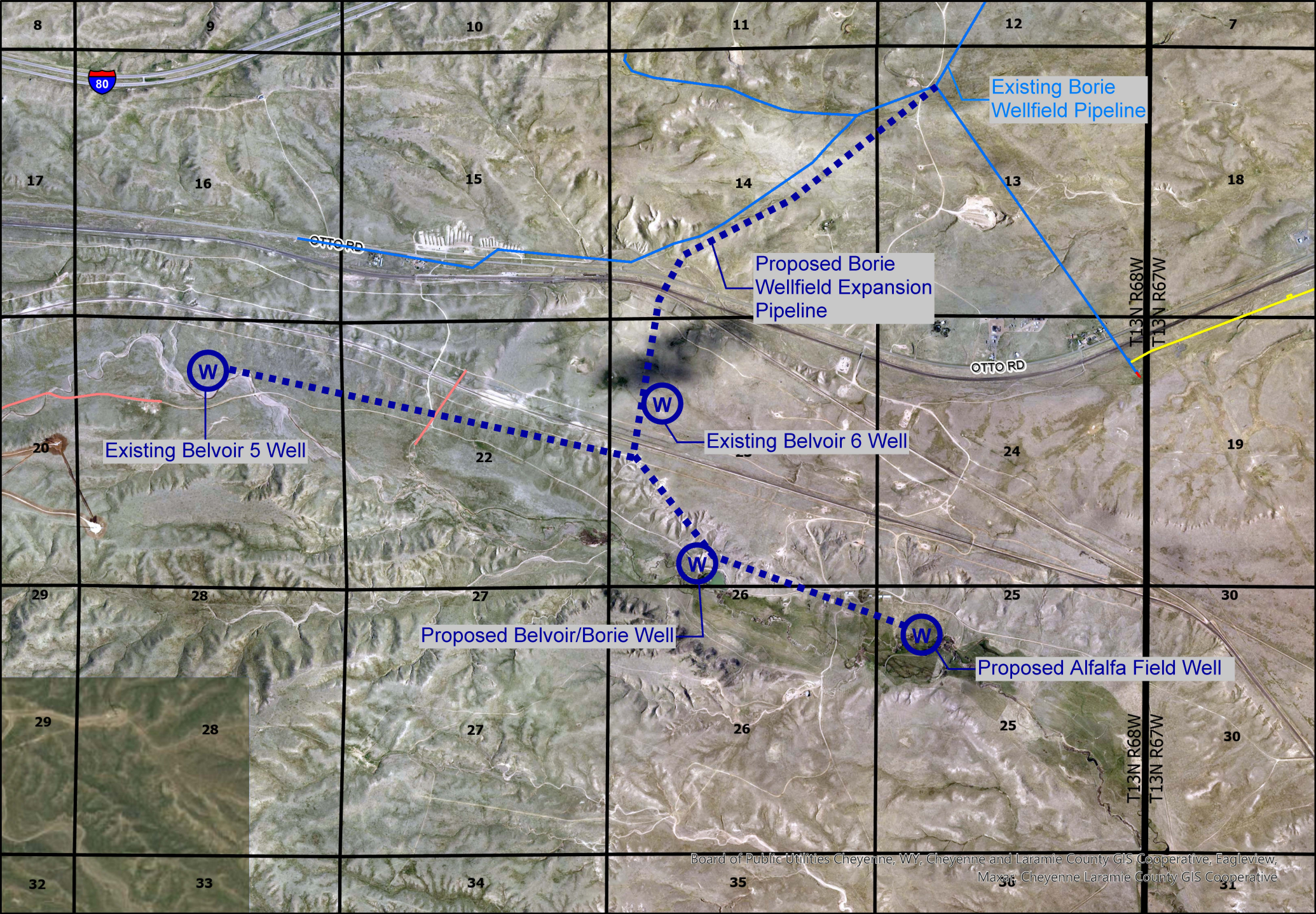
Sponsor cost share will be \$400,000 from system development fees as part of Fiscal
Year 2024 budget. Funding to finance subsequent production and transmission work will
be through applications to WWDC Level III Funding, WaterSMART grant program, and
DWSRF loan.

5. Cost Estimates

Use the following format in calculating project cost estimates. (An alternative format may be prepared and submitted, but it must be clearly itemized and show total project cost for determining WWDC grant).

A. Preparation of Hydro-Geologic Analysis and/or Well Siting Study:	<u>Currently under contract - \$0</u>
B. Permitting:	<u>Test wells - \$0</u>
C. Advertising, Contractor Procurement, Contracts:	<u>\$80,000 - P.G.</u>
D. Well Construction [1]:	<u>\$600,000</u>
E. Aquifer (Pump) Testing:	<u>\$80,000</u>
F. Water Quality Analysis:	<u>\$30,000</u>
G. Project Management/Subcontracts:	<u>In-house - \$0</u>
H. Miscellaneous Applications/Procurements (Specify):	<u>\$0</u>
I. Reports:	<u>\$10,000 - P.G.</u>
J. Total Project Cost:	<u>\$800,000</u>
K. Rounded:	<u>\$800,000</u>

[1] Including, but not limited to: Mobilization, Bonds & Insurance, Drill-Furnish-Install Casing/Screen, Furnish-Install Filter Pack/Seal, Rig Time, Standby Time, Geophysical/Video Log, Development, Disinfection, Plugging and Abandonment, De-Mobilization, Reclamation. Any inflation costs, as determined by the WWDC, will be applied to the Total Project Cost.



Board of Public Utilities Cheyenne, WY, Cheyenne and Laramie County GIS Cooperative, Eagleview, Maxar Cheyenne Laramie County GIS Cooperative



Borie Wellfield Expansion



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